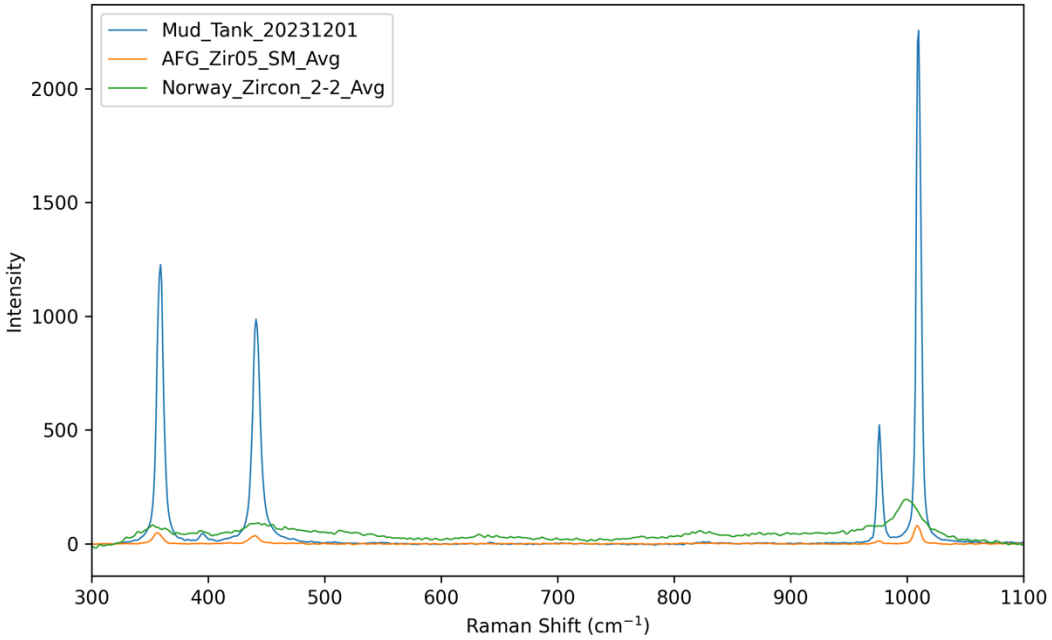
Stable isotope dilution approach to measuring mineral replacement rates at high P-T conditions: Replacement of metamict zircon by hydrothermal zircon

John C. Ayers, Pan Hu, and Blake Wallrich, Vanderbilt University
Chen Zhu, Indiana University Bloomington
Keewook Yi, Korea Basic Science Institute

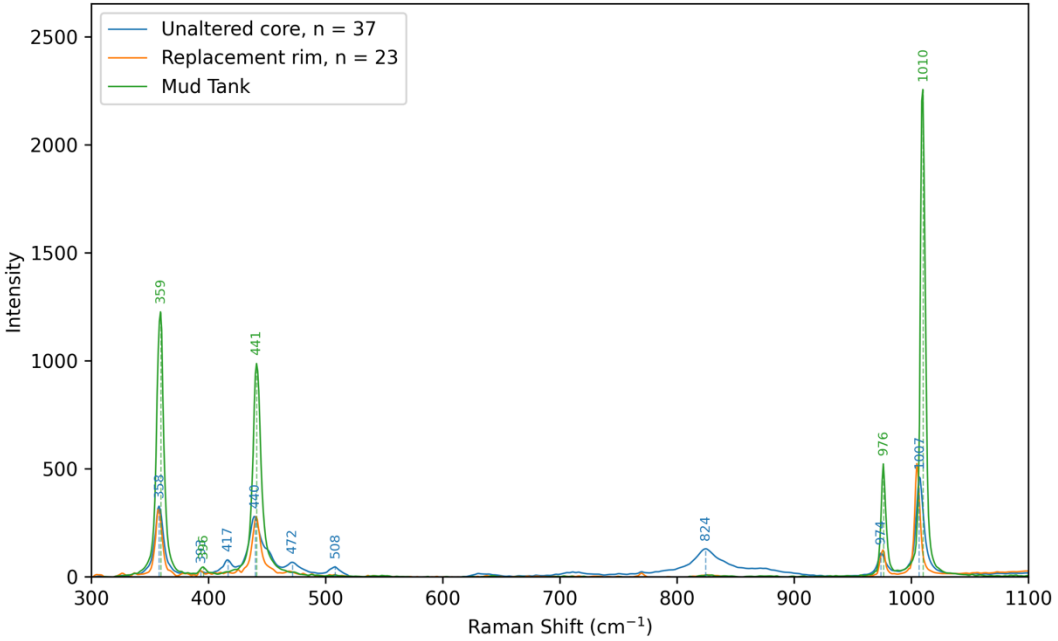
VANDERBILT UNIVERSITY

Raman Spectra

Starting material: AFG zircon



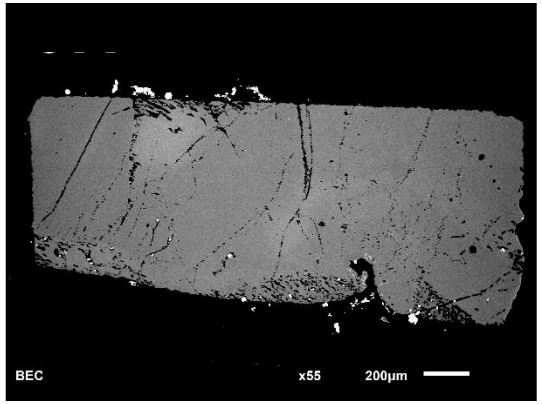
Zir016 run products

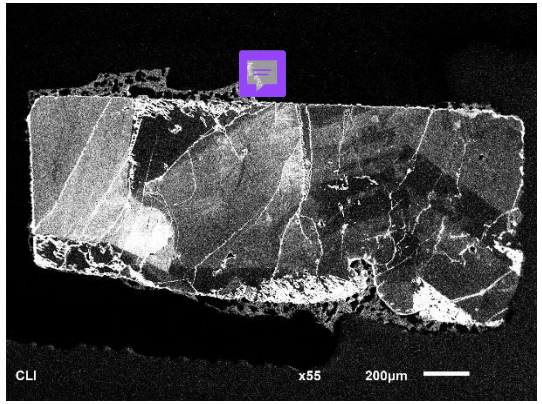


BSE

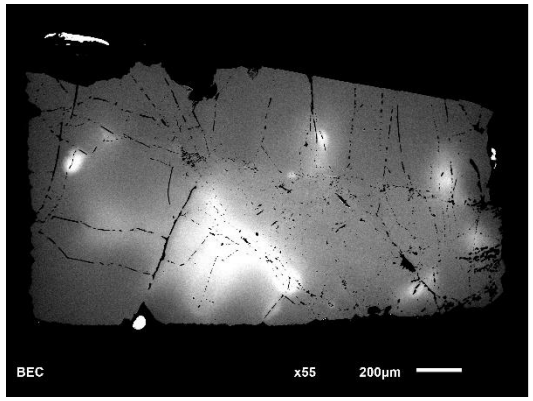
CL

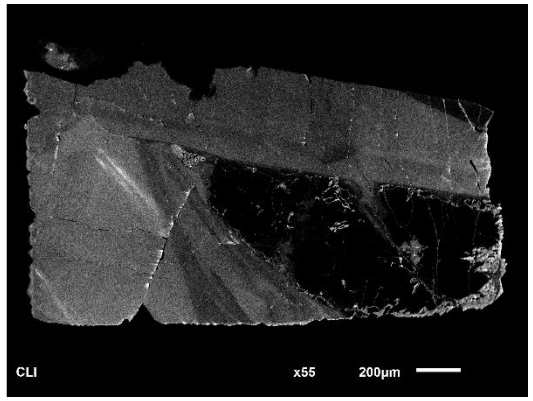
3 days



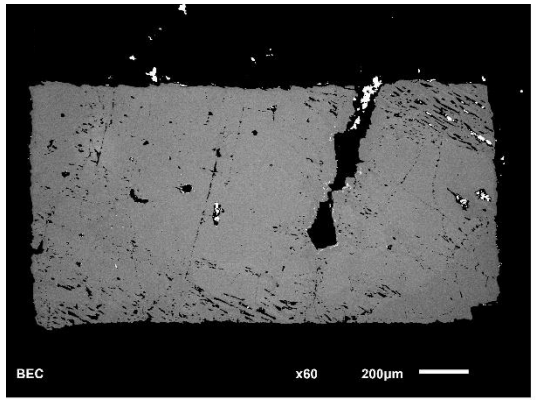


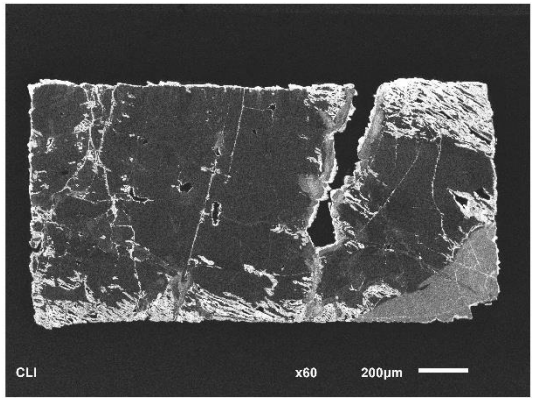
7 days





19 days



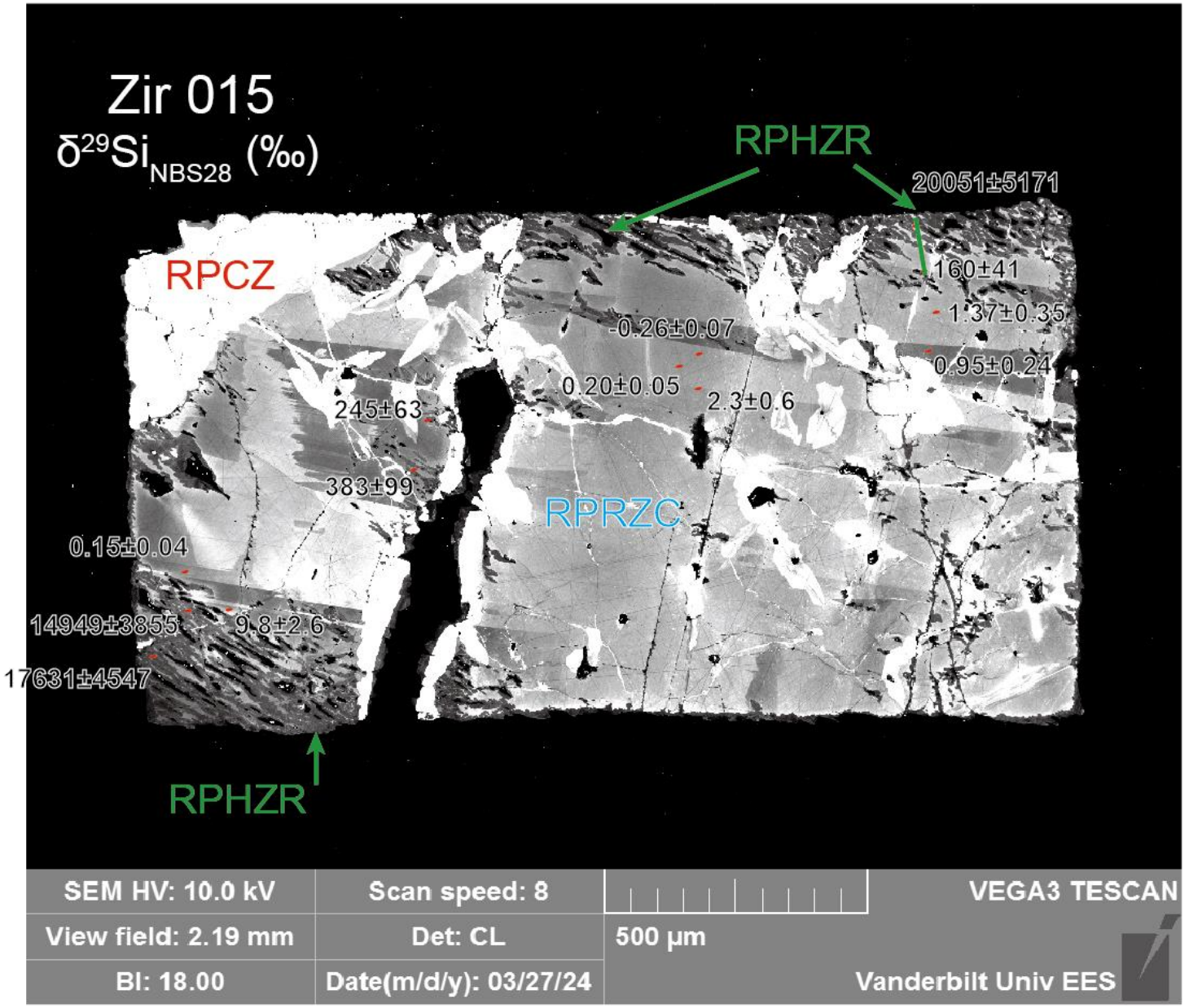


The replacement occurs on mineral surface and cracks;

The replacement domain show porous texture;

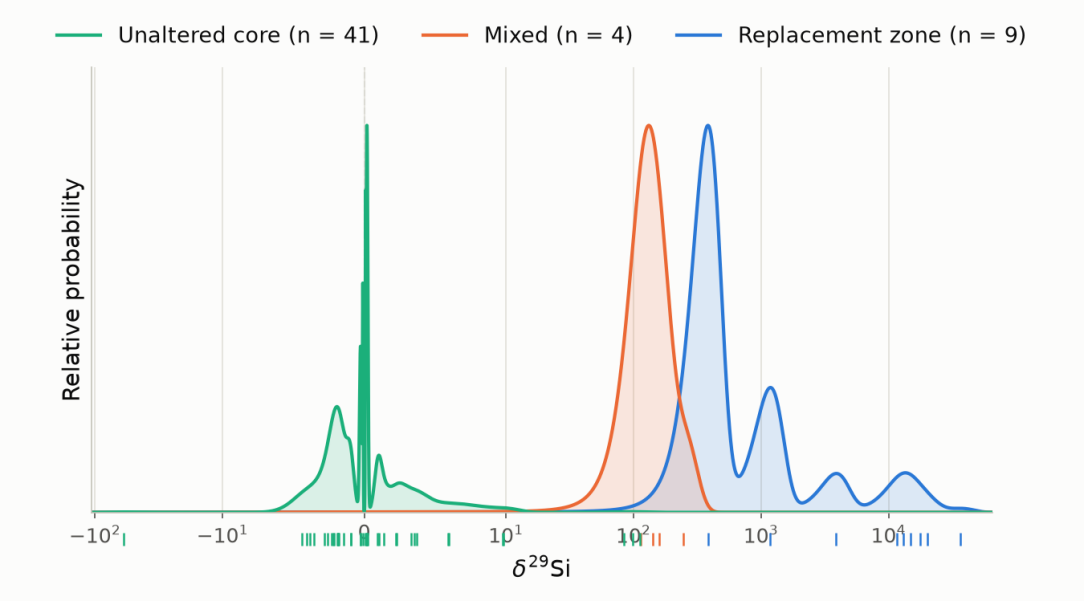
Increase in the percentage of replacement through time.

Run product Silicon isotope compositions

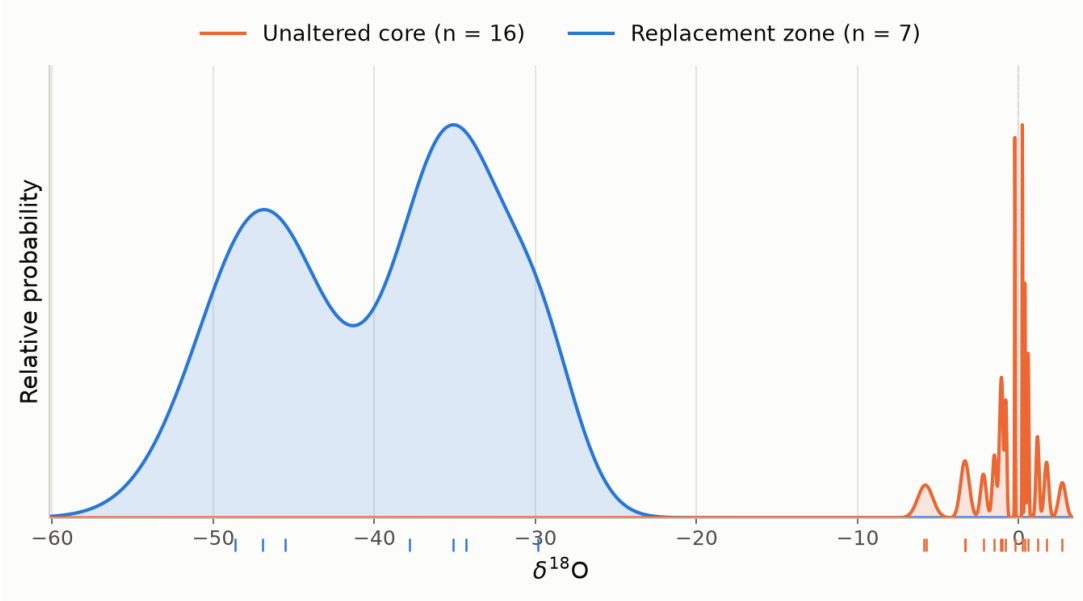


SIMS isotope spot analyses

Silicon isotope kde

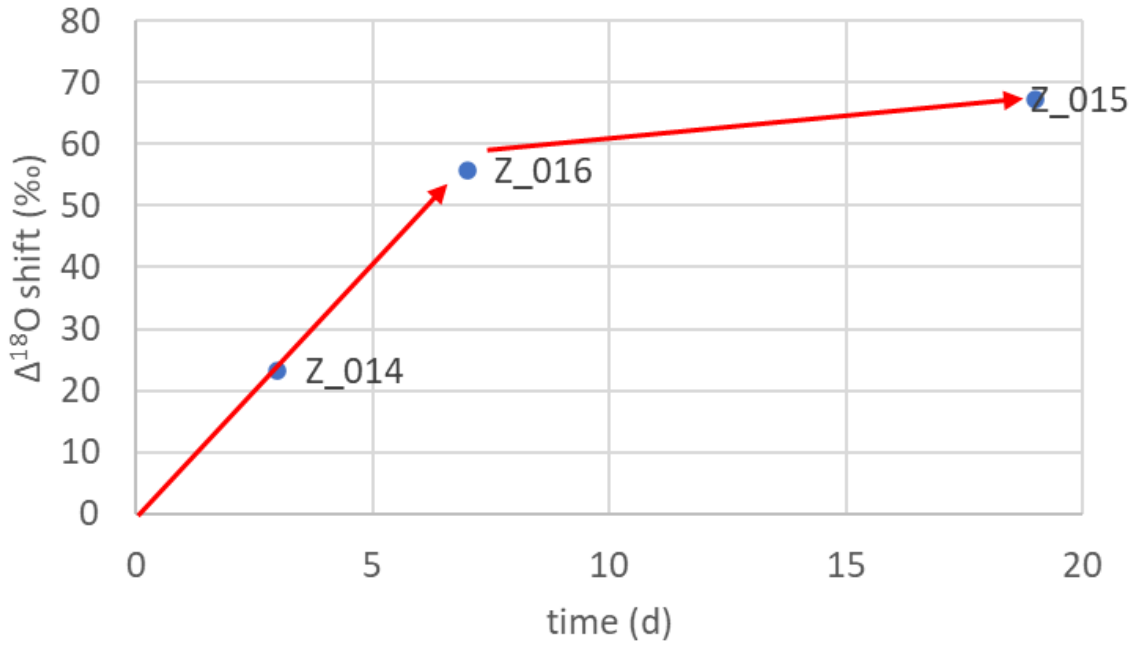


Oxygen isotope kde

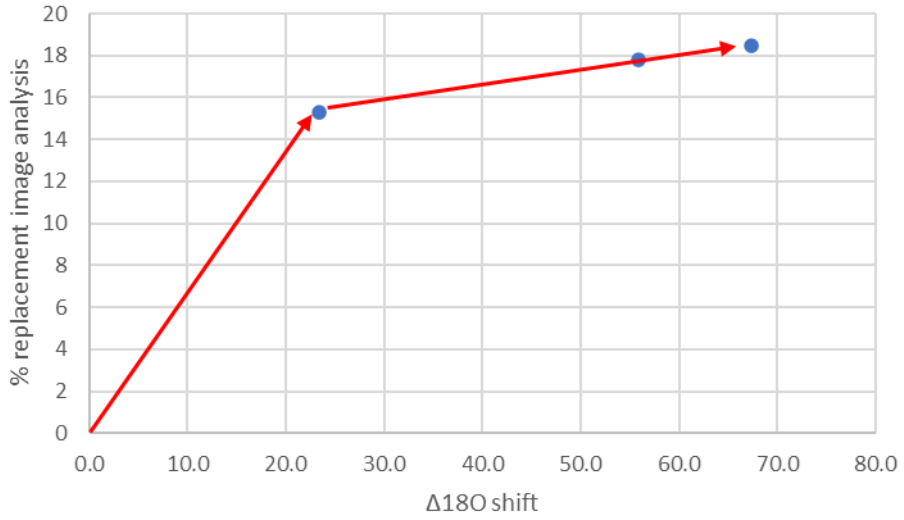


Oxygen isotope results from capsule fluid

Reaction rate is high and then levels off

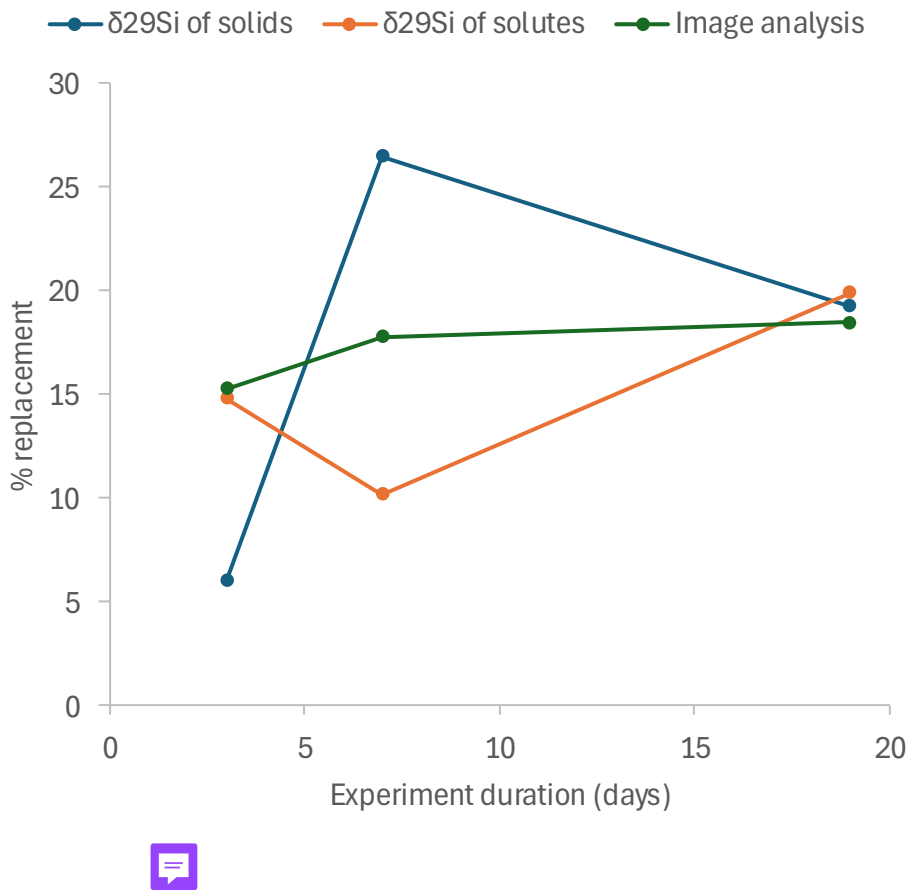


Two measures of extent of reaction show good agreement



Percent Replacement Results for AFG at 800°C

Duration (days)	δ29Si of solids	δ29Si of solut	Image analysis
3	6.0	14.8	15.3
7	26.5	10.2	17.8
19	19.3	19.9	18.5



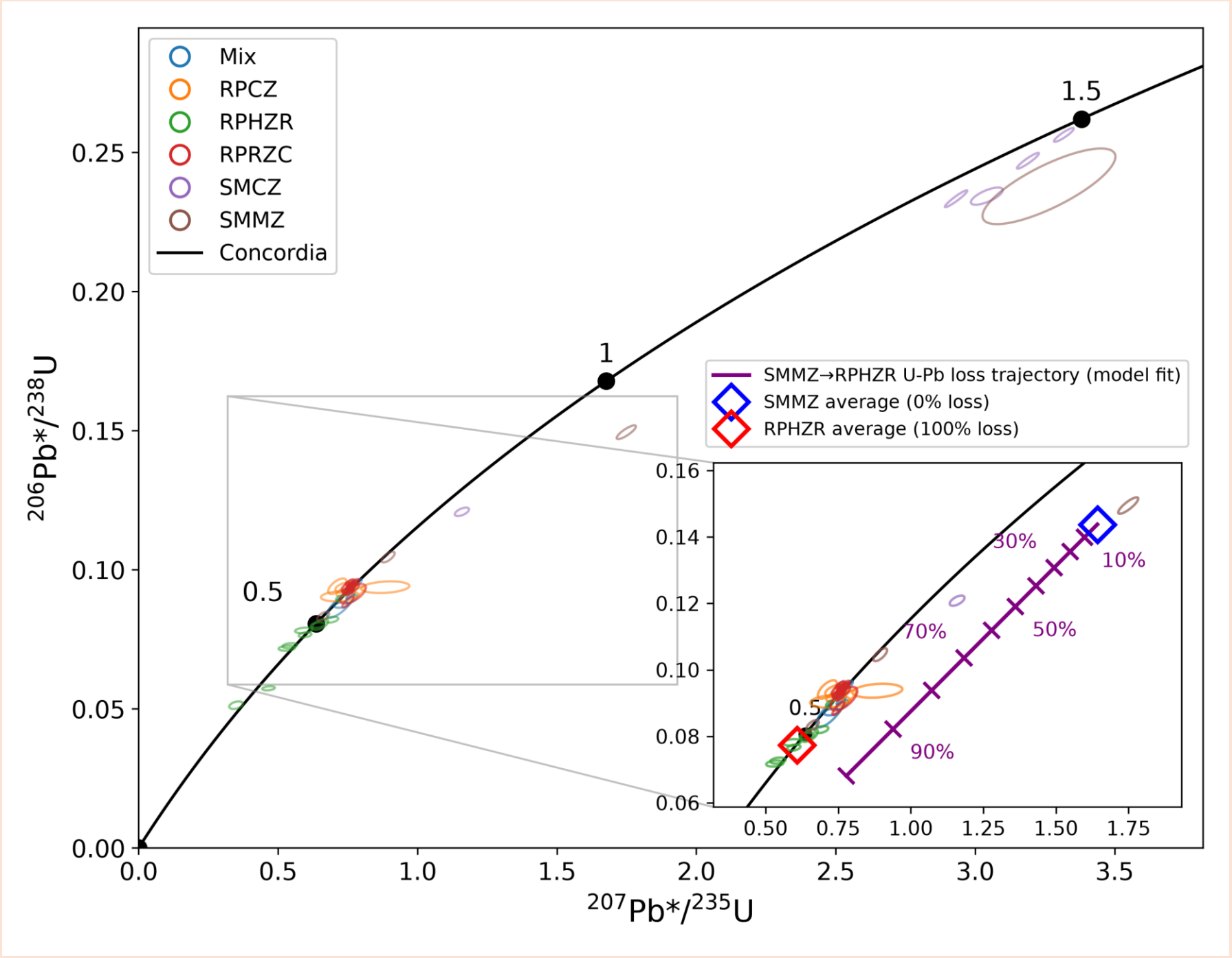
- Calculation of % replacement from average $\delta^{29}Si$ assumes a homogeneous (batch equilibrium) model.
- In experiments Zir-017 and Zir-018 Norway zircon was completely replaced after only one day at 600°C.

Replacement rates

Material	AFG	AFG	AFG	AFG	AFG	AFG	Norway	Norway
Method	Solids	Solutes	Solids	Solutes	Solids	Solutes	Solutes	Solutes
duration (s)	259200	259200	1641600	1641600	604800	604800	86400	86400
replaced $ZrSiO_4$ (moles)	9.4E-05	1.5E-05	3.3E-05	3.8E-05	3.9E-05	1.5E-05	2.2E+00	1.7E-01
Bulk rate (moles kg $H2O^{-1}$ s^{-1})	3.0E-06	4.7E-07	1.7E-07	1.9E-07	5.4E-07	2.1E-07	2.1E-01	1.7E-02

- Replacement rate much higher for Norway due to higher alpha dose and higher surface area.
- Norway was completely replaced, so rate estimate is a minimum.

SIMS U-Pb geochronology results



Conclusions

- Hydrothermal fluids can cause replacement of radiation-damaged zircon by crystalline hydrothermal zircon.
- A new stable isotope method was used to measure the rate of replacement.
- Replacement rate increases as degree of radiation damage increases.
- Hydrothermal zircon may form through replacement of metamict $ZrSiO_4$.
- Replacement zone U-Pb age = timing of fluid event.
- Future work: use SIMS to measure oxygen isotope composition of quenched solutes and do depth profiling on thin rims.